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Research Paper

Genome-wide identification and analyses of the AHL gene family in banana (*Musa acuminata*) and the functional analysis of *MaAHL43* in fruit ripening

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ABSTRACT

Banana (*Musa acuminata*) is one of the most important and popular fruit crops worldwide. The AT-hook motif nuclear localized (AHL) is a conserved transcription factor family involved in the regulation of plant growth and development and stress response. However, the information of *MaAHL* genes remain largely elusive. In this study, a total of 48 *MaAHL* genes were identified based on *Masu acuminata* genome, which were distributed unevenly across 11 chromosomes. Phylogenetic analysis showed that the MaAHL proteins could be divided into two clades of A and B and subdivided into three types. The *cis*-regulatory elements in the promoter regions of *MaAHL* genes were mainly involved in diverse functions including plant growth and development, phytohormones responses, and stress response process. Expression profile analysis showed that *MaAHL* genes responded to osmotic stress and *Fusarium oxysporum* infection. In addition, the expression of clade-A *MaAHL* genes were generally downregulated while clade-B *MaAHL* genes were up-regulated during fruit ripening in banana. RT-qPCR showed that the expression of *MaAHL43* gene was significantly up-regulated in middle and late stages of fruit ripening. MaAHL43 is located in nuclear and has no transactivation activity in yeast. Overexpression of *MaAHL43* gene could promote banana fruit ripening, indicating MaAHL43 might a regulator of fruit ripening in banana. Taken together, this study would provide valuable information for further studies on the functions of *MaAHL* genes, and providing a new target for further research on the regulation network of banana fruit ripening.

1. Introduction

The AT-hook motif nuclear localized (AHL) is a conserved transcription factor family involved in plant growth and development, organ construction, abiotic stress tolerance and immune response. Members of the *AHL* gene family have been identified in many plants (Zhao et al., 2013; Bishop et al., 2019; Zhao et al., 2020; Kumar et al., 2023; Wang et al., 2023; Chen et al., 2024). The AHL proteins contains one or two AT-hook motif(s) which required for DNA binding, and a plants and prokaryotes conserved (PPC/DUF296) domain at C-terminal, involved in nucleus localization and protein interaction (Aravind et al., 1998; Zhao et al., 2014). The four amino acid residues Arg(R)-Gly(G)-Arg (R)-Pro(P) as the core of AT-hook motif. The AHL proteins could be divided into three types according to the amino acid sequence of the C-terminal of the RGRP core, leading to differences in the ability of affinity to bind to DNA (Aravind et al., 1998; Fujimoto et al., 2004; Gordon et al., 2010). The PPC/DUF296 domain is about 120 amino acids in length, and its secondary structure and function are highly conserved in eukaryotes/prokaryotes (Fujimoto et al., 2004; Zhao et al., 2014).

The AHL family plays a critical role in plant growth and development. AHL22 acts as a chromatin remodeling factor that modifies the architecture of FLOWERING LOCUS T chromatin by modulating both histone 3 acetylation and methylation, thereby regulating flowering time in *Arabidopsis* (Xiao et al., 2009; Yun et al., 2012). Ectopic expression of *AHL15* suppresses AM maturation and promotes longevity in monocarpic *Arabidopsis* and tobacco (Karami et al., 2020). AHL15 promotes secondary growth and branching in *Arabidopsis* inflorescence stems (Rahimi et al., 2022a). *DP1*(*DEPRESSED PALEA 1*) encodes an AT-hook protein and affect sterile lemma identity on the palea side and floral organ number in rice (Jin et al., 2011). GhAT1 assists in the specification of fiber cells by repressing *FSltp4* gene in the non-fiber tissues of the cotton plant (Delaney et al., 2007). The expression of *Prupe.1G530300.1 and Prupe.1G034400.1* (two *AHL* genes) is positively correlated with total sugar or sucrose content (Mollah et al., 2022), and

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transient overexpression of *Prupe.1G530300.1* and *Prupe.1G034400.1* had positive effect on sucrose and total sugar accumulation in peach fruits (Zhao et al., 2023). In addition, AHL proteins also participated in response processes of abiotic stress and plant immune. Overexpression of *OsAHL1* enhanced multiple stress tolerances in rice plants during both seedling and panicle development stages (Zhou et al., 2016). The expression of clade A *SlAHL* genes were distinctly changed under abiotic stresses, overexpression of *SlAHL1* and *SlAHL7* responded drastically upon ABA, salt and drought stresses (Wang et al., 2023). Expression of *CaATL1* was specifically induced in host- and non-host-resistant responses against bacterial and viral pathogens in pepper, and overexpression of *CaATL1* in tomato enhanced disease resistance against bacterial and oomycete pathogens (Kim et al., 2007). OsATH1 specifically bound to the promoter of class III peroxidases *OsPrx30* gene to regulated rice bacterial blight-induced ROS accumulation (Liu et al., 2021).

Banana (*Musa acuminata*) is one of the most important and popular climacteric fruit crops worldwide. The fruit ripening is regulated by genetic, environmental and physiological factors, among which ethylene plays a critical role in this process (Kou et al., 2021). In recent years, several regulatory factor of banana fruit ripening have been identified, such as *MaERF9* (Xiao et al., 2013), *MaERF11* (Han et al., 2016), *MaDEAR1* (Fan et al., 2016), *MaERF012* (Chen et al., 2022), *MaMYB16L* (Jiang et al., 2021), *MaDOF23* (Feng et al., 2016), *MaNAC083* and *MaMADS1* (Wei et al., 2023). However, the ripening process of banana fruit is governed by a complex regulatory network, and the regulatory factors in this network remain to be elucidated. The *AHL* gene family was widely participated in the regulation of plant growth and development and stress response, but the information of *MaAHL* genes have not yet been characterized. Given the distinct roles of *AHL* family members in diverse biological processes, the characterization of *AHL* genes in banana would facilitate to understand their functions and lay the foundation for future application in the genetic improvement of banana. In this study, a total of 48 *MaAHL* genes were identified based on *Masu acuminata* genome. The chromosomal location, genetic structure, phylogenetic relationship and expression profiles of *MaAHL* gene family were investigated. Moreover, it was found that *MaAHL43* might participate in fruit ripening process. These findings would provide valuable information for further studies on the functions of *MaAHL* genes, and facilitate clarification of molecular regulatory network of ethylene mediated fruit ripening in banana.

2. Result

2.1. Genome-wide characterization of AHL genes in Musa acuminata

To identify the candidate *MaAHL* genes, the sequences of 29 *Arabidopsis* AHL proteins were used to search through BLAST based on the *M. acuminata* genome database. Subsequently, the retrieved putative genes were filtered through SMART and NCBI Batch CD search to remove proteins with incomplete domains. A total of 48 *MaAHL* genes were obtained and named according to their position on the chromosome (Table S1). The MaAHL proteins have a peptide length ranging from 266 (MaAHL1 and MaAHL47) to 605 aa (MaAHL31), and the molecular weight ranges from 26.99 (MaAHL1) to 66.17 (MaAHL31) kDa. In addition, the predicted isoelectric point (pI) of MaAHL members were between 5.31 (MaAHL47) ~ 10.08 (MaAHL27), and there is an equal proportion of acidic and alkaline proteins in all MaAHL proteins. The results of instability index analysis showed that the instability index of MaAHL members ranges from 31.86 (MaAHL32) to 71.43 (MaAHL27), and almost MaAHL proteins except MaAHL32 and MaAHL7 were structurally unstable (instability index > 40). Moreover, there were only three hydrophobic proteins (MaAHL3, MaAHL14 and MaAHL16) in MaAHL members, and the rest were hydrophilic proteins. Prediction of subcellular localization results suggested that MaAHL members were mainly located in nucleus, followed by the chloroplast and cell membrane.

Chromosomal location analysis showed that 48 *MaAHL* genes were distributed unevenly on the 11 chromosomes (Chr) (Fig. 1). Chr03 and Chr05 harbored the highest gene number of *MaAHL* genes (7 genes), followed by Chr01 and Chr06 (5 genes), and the lowest number was on Chr04, with only one *MaAHL* gene. In addition, several *MaAHL* genes were located on Chr01, 02, 05, and 06 in adjacent positions, indicating a replication event during evolution.

2.2. Phylogenetic, conserved motif and gene structure analysis of MaAHL genes

Phylogenetic analysis showed that the AtAHL and MaAHL proteins could be divided into two clades of A and B, which contained 19 and 29 MaAHL family members, respectively (Fig. 2). Clade-A and clade-B could be further subdivided into four and five subgroups, respectively. The AHL proteins of *Arabidopsis* and banana in each branch cluster separately, indicating that the AT-hook gene family has greatly differentiated in evolution. To investigate the structural characteristics of MaAHL proteins, ten conserved motifs were identified through MEME database search. All MaAHL members contain a conserved DUF296/PPC domain and could be classified into three types based on the number and composition of the AT-hook motif (Fig. 3A and B). A total of 29 type I MaAHL members have an AT-hook1 motif (motif3) which shared the same conserved R-G-R-P-X-G core (Fig. 3D and E). There is only one type II MaAHL member (MaAHL15) was identified, which harbored the AT-hook2 motif (motif5) with a R-G-R-P-R-K core. Eighteen type III MaAHL members have both AT-hook1 and AT-hook2 motifs. Motif 1, 4, and 2 were found in all MaAHL members and might be a crucial part of DUF296/PPC domain.

The gene structure of MaAHL family members were further investigated (Fig. 3C). The results showed that the sequence length of *MaAHL* genes varied greatly, with the longest (*MaAHL31*) DNA sequence length approaching 10 kb and the shortest (*MaAHL47*) less than 1 kb. The number distribution of exon-intron structure of *MaAHL* genes also varied greatly, with exon number ranging from 1 to 9. The *MaAHL* genes located on the same branch exhibited similar structures. The most type I *MaAHL* genes have only one exon, the other four type I *MaAHL* genes (*MaAHL8, MaAHL13, MaAHL35* and *MaAHL36*) have two exons except for *MaAHL31*. Interestingly, *MaAHL31* gene belongs to type I *MaAHL* genes, but it has the longest DNA sequence length and the highest exon number, which is due to the presence of fatty acid hydroxylase (FA_ hydroxylase) domain at the N-terminal of the MaAHL31 protein (Fig. 3B). Compare with type I MaAHL members, all type II and type III members have UTR region and longer DNA sequence length, exhibiting multiple-exon structure, this indicating that *MaAHL* genes might diverge functionally during evolution.

2.3. Cis-element analysis of the promotors of MaAHL genes

To further investigate the potential biological processes and signaling pathways involved in the *MaAHL* genes, the 2 kb upstream presumptive promoter sequences were employed to analyze the characterization of *cis*-element of *MaAHL* genes. The result showed that the promoter regions of *MaAHL* genes were enriched with a series of *cis*-elements involved in plant growth and development, phytohormone responsiveness and stress responses (Fig. 4). The light responsive elements, including TGA-element, TCT-motif, AT1-motif, TCT-motif, GT1-motif, ACE and 3-AF1 binding site, were identified in the promoter region of all MaAHL members, indicating *MaAHL* genes might be a vital part of light-mediated plant growth and metabolism process. Approximately 54.2% and 16.7% of *MaAHL* genes contained elements involved in meristem expression and seed-specific regulation, respectively, suggesting *MaAHL* genes might participate in regulation of meristem and organ development. Meanwhile, abscisic acid response elements (ABREs) and *cis*-acting regulatory elements involved in the MeJA-

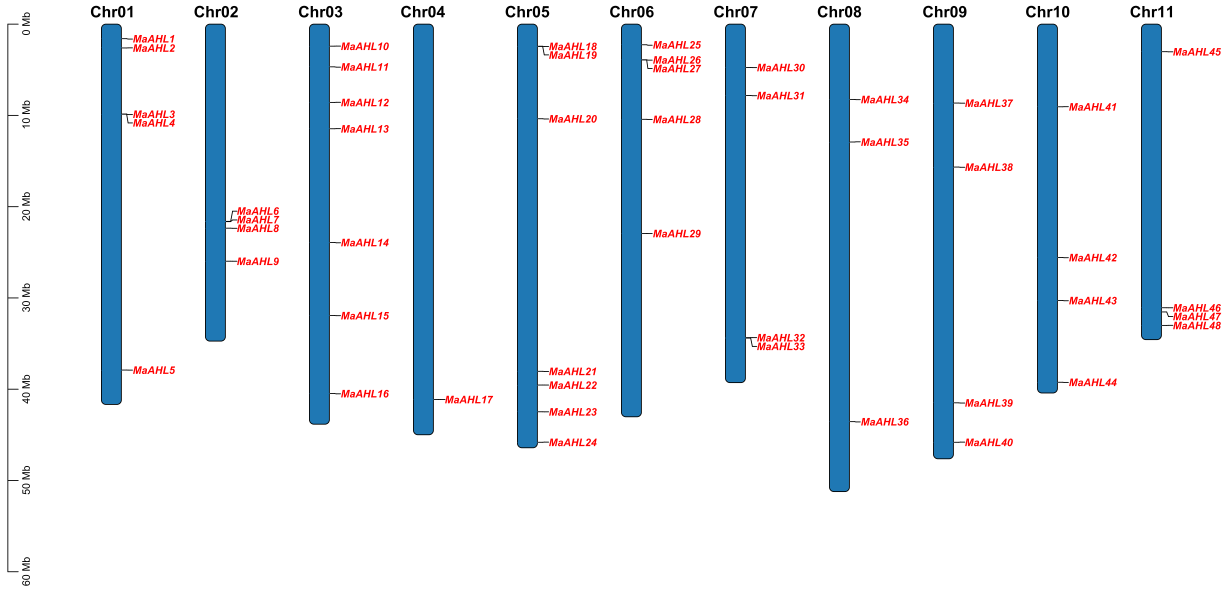

Fig. 1. Chromosomal locations for the *MaAHL* genes on 11 chromosomes. The chromosome number is represented at each bar top. Mb, megabase.

responsiveness (TGACG-motif and CGTCA-motif) were found in most of *MaAHL* genes. The auxin-responsive (element TGA-element) was identified in 28 *MaAHL* genes, while two gibberellin-responsive elements (GARE-motif and P-box) and two salicylic acid response elements (TCA-element and SARE) were distributed in 22 *MaAHL* genes. Moreover, approximately 50.0%, 58.3%, and 37.5% of *MaAHL* genes contained low-temperature elements (LTR), drought induction elements (MBS) and defense and stress responsiveness (TC-rich repeats), respectively. These results indicate that the *MaAHL* genes could be widely participated in diverse functions including plant growth and development, phytohormones responses, and stress response process.

2.4. *Expression **patterns** of MaAHL genes under a**biotic and biotic stress***

The expression profiles of *MaAHL* genes under osmotic stress were analyzed based on public transcriptome date (Fig. 5). The result showed that many *MaAHL* genes exhibited a significant change in expression under osmotic stress. The expression levels of *MaAHL8* and *MaAHL10* were significantly up-regulated, while *MaAHL6* and *MaAHL9* were significantly down-regulated in root of three banana genotypes under osmotic stress. The expression of *MaAHL2 and MaAHL42* were significantly induced in ‘Mbwazirume’ (AAA) but suppressed in ‘Cachaco’ (ABB) and ‘GrandeNaine’ (AAA) under osmotic stress, while the opposite was observed in *MaAHL23*.

The expression profiles of *MaAHL* genes response to *Fusarium oxysporum* f.sp. *cubense* race1 (R1) and tropical race4 (TR4) were further explored (Fig. 5). The expression levels of *MaAHL2* and *MaAHL4* in root were up-regulated gradually after R1 and TR4 infection, while the expression of *MaAHL9* was increasing first and then decreasing in root and corm. *MaAHL23* gene also exhibits an initial increase followed by a decrease trend in root and corm after R1 infection, but its expression in root and corm down-regulated gradually after TR4 infection. Noteworthy, the expression of *MaAHL34* were up-regulated gradually in both root and corm with the extension of the R1 and TR4 infection time, suggesting it might play a vital role in plant immune response. These results indicating *MaAHL*-mediated biological processes might be easily affected by external environmental condition in banana.

2.5. *Expression **of** MaAHL genes during fruit ripening*

To investigate the potential biological function of *MaAHL* genes during fruit ripening, three fruits from different periods were selected for RNA-seq to analyze the expression patterns of *MaAHL* genes. The basic information for the transcriptome was shown in Table S2. The result showed that a total of 19 *MaAHL* genes were expressed (FPKM > 1) in fruit, and most of those *MaAHL* genes exhibited a down-regulated expression trend during fruit ripening (Fig. 6A). The expression of four genes, including *MaAHL28, MaAHL31, MaAHL35* and *MaAHL43*, were up-regulated during fruit ripening. In addition, *MaAHL* genes on the same branch showed similar expression patterns. For example, *MaAHL28* and *MaAHL43* were both expressed at a high level in late stage of fruit ripening, and the expression of *MaAHL23* and *MaAHL27* were down-regulated gradually. *MaAHL22, MaAHL34, MaAHL45* and *MaAHL46* genes with close phylogenetic relationships and exhibit consistent expression patterns. To verify the reliability of transcriptome data, the expression pattern of 15 *MaAHL* genes were evaluated by RT-qPCR analysis. As expect, the expression trends of these *MaAHL* genes were basically in accordance with the transcript abundance changes from RNA-seq data (Fig. 6B). Noteworthy, both RNA-seq and RT-qPCR results showed that the expression of *MaAHL43* gene was significantly up-regulated in middle and late stages of fruit ripening, indicating that *MaAHL43* gene might positively regulate the ripening process of banana fruit. Therefore, *MaAHL43* gene was selected as a candidate on banana fruit ripening for further investigation.

2.6. *MaAHL43 was involved in banana fruit ripening*

To characterize the subcellular localization of MaAHL43, *35S::MaAHL43-GFP* plasmid was constructed and transiently overexpressed in tobacco leaves. The GFP signal was mainly localized in the nucleus as confirmed by a mCherry-labelled nuclear marker, indicating that the MaAHL43 is a nuclear localization protein (Fig. 7A). Subsequently, MaAHL43-pGBKT7 plasmid was constructed and introduced into yeast strain Y2HGold to investigate the transcriptional activation activity of MaAHL43. The result showed that the yeast cells transformed with MaAHL43-pGBKT7 were able to grow well on SD-Trp, but could not grow on SD-Trp/His/Ade medium, indicating MaAHL43 has no trans-activation activity in yeast (Fig. 7B). To further investigate the function of *MaAHL43* gene in banana fruit ripening, *35S::MaAHL43* vector was

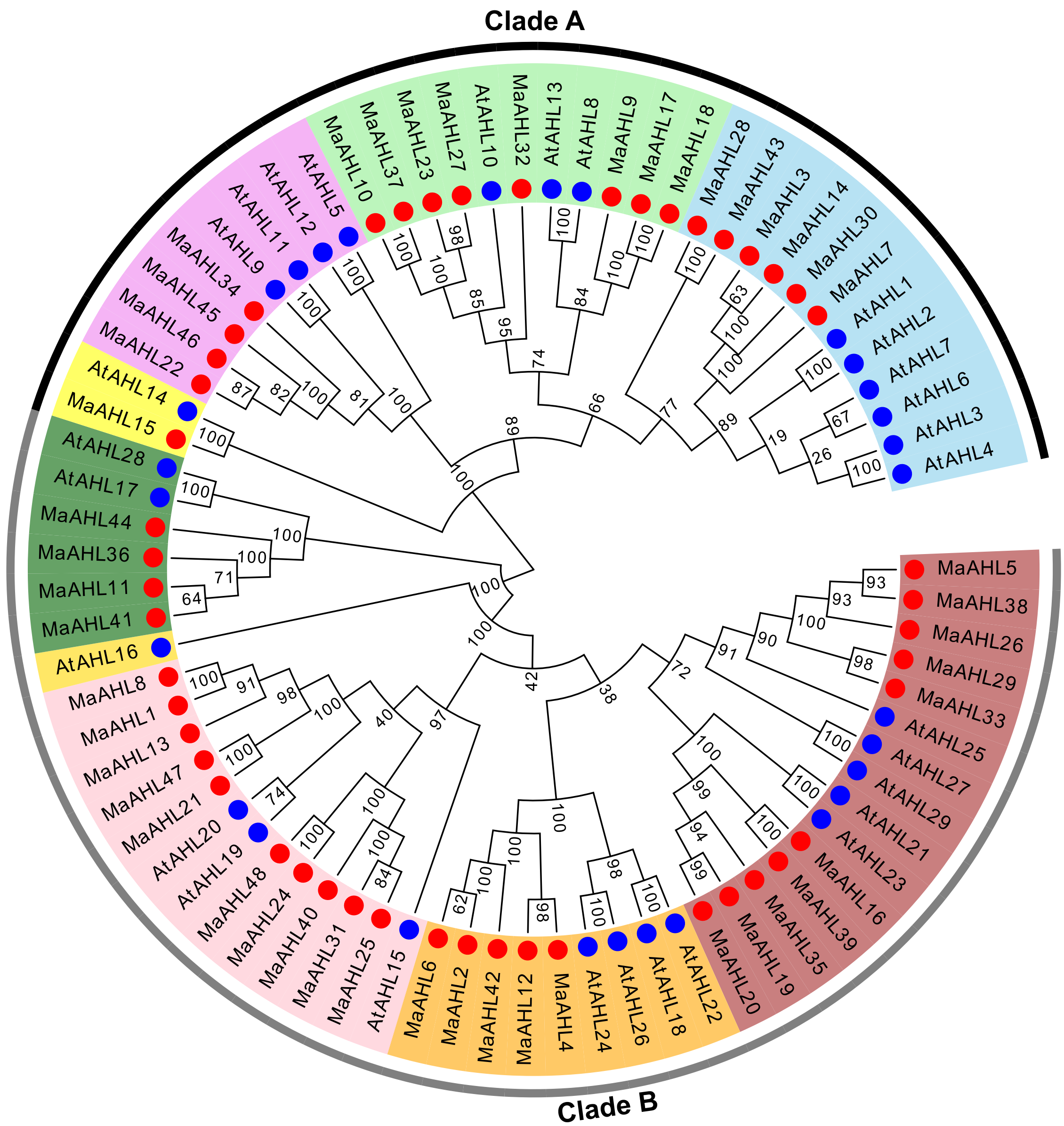

Fig. 2. Phylogenetic analysis of AHL protein involved 29 Arabidopsis AHL protein and 48 banana AHL protein sequences. Blue and red circles represent the AtAHL proteins and radish MaAHL proteins, respectively. The clade A and clade B group is marked in black and grey, respectively.

constructed and transiently expressed in unripe banana fruit (Fig. 7C). Compared with control, the expression of *MaAHL43* gene was highly upregulated in fruits overexpressed *MaAHL43* gene (Fig. 7D). The fruits overexpressed *MaAHL43* gene showed accelerated yellowing than in the control fruits (Fig. 7E). Furthermore, the expression profiles of several ethylene pathway related genes were detected in control and *MaAHL43*-OE fruits during ripening though RT-qPCR. The result showed that the expression peaks of *MaACS1, MaACS12, MaACO1, MaACO4, MaACO5*, and *MaACO8* gene was significantly advanced in *MaAHL43*-overexpressing fruit during ripening, indicating that overexpression of *MaAHL43* gene activates the expression of genes related to ethylene pathway and promotes fruit ripening. These results suggested that *MaAHL43* gene might a positive regulator of fruit ripening in banana.

3. Discussion

The *AHL* family genes have been identified in many sequenced plants, including *Arabidopsis* (Zhao et al., 2013), rice (Kumar et al., 2023), cotton (Zhao et al., 2020), maize (Bishop et al., 2019), tomato (Wang et al., 2023) and radish (Chen et al., 2024). Here, a total of 48 *MaAHL* genes were identified from *M. acuminata* genome and their characteristics were investigated (Table S1), which is the first systematic characterization of AHL gene family in banana. The number of *AHL* genes in banana was obvious increase than that in *Arabidopsis* (29) and rice (26), which may be caused by the extensive gene diversification of *MaAHL* genes after three whole-genome duplication (WGD) event (D'Hont et al., 2012). The AtAHL and MaAHL proteins cluster almost separately in each branch of phylogenetic trees, indicating that the AT-hook gene family has greatly differentiation in evolution. As previously reported for various other plants (Churchill and Travers, 1991; Huth et al., 1997; Reddy et al., 2005), all MaAHL proteins contain a conserved DUF296/PPC domain and one or two AT-hook motif(s) (Fig. 3B). Additionally, all 48 MaAHL members could be classified into three types based on AT-hook motif, and members of the same type are similar in gene structure (Fig. 3A). Noteworthy, there is only one type II MaAHL member (MaAHL15) was identified in banana, suggesting the AT-hook 1 motif (motif 3) might be a critical component for the function of *MaAHL* genes. Compare with type II and type III, type I *MaAHL* genes

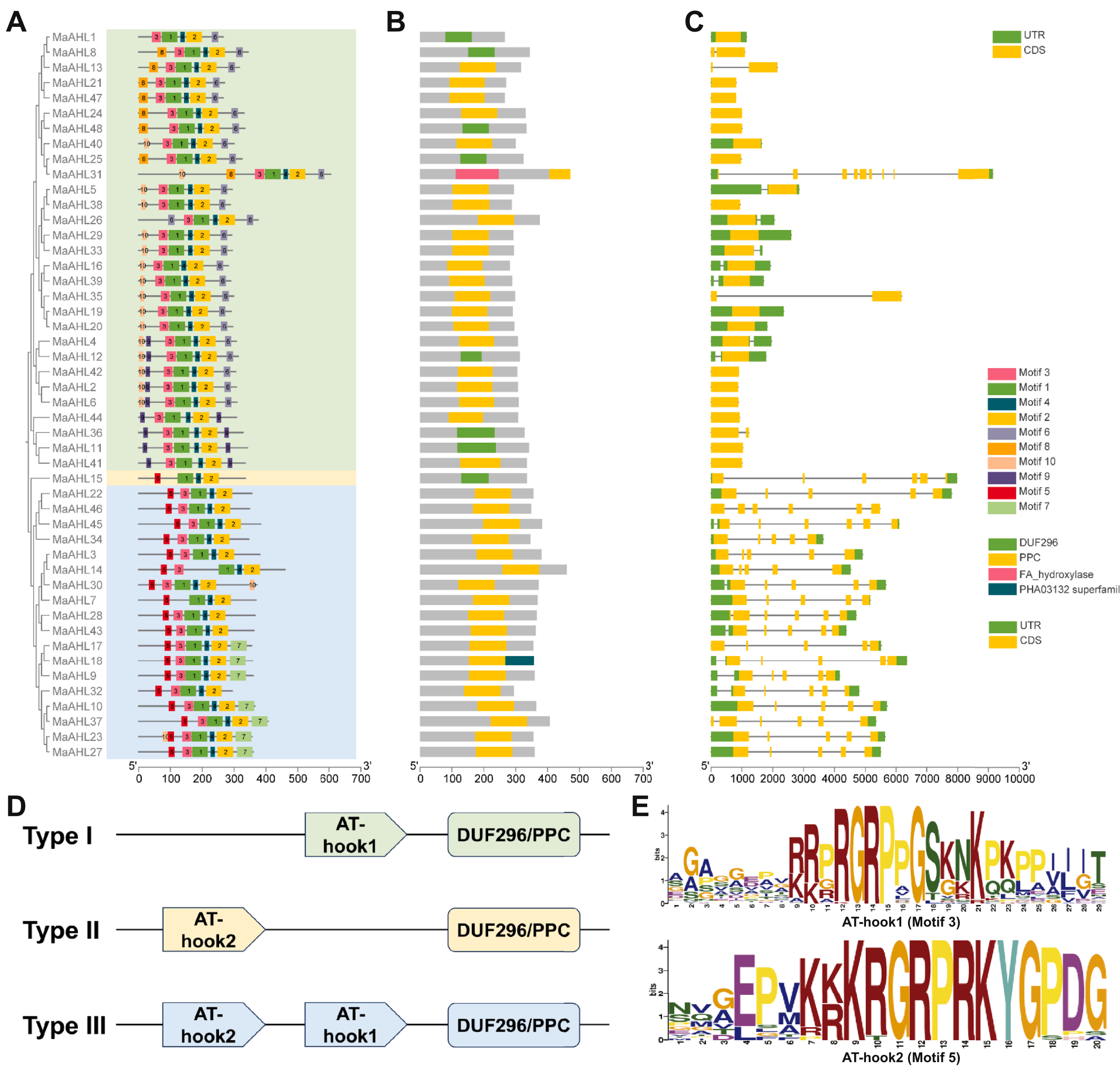

Fig. 3. Phylogenetic analysis, motif distribution, conserved domains and gene structure of *MaAHL* genes. (A) Phylogenetic tree using the neighbor-joining method, and the conserved motifs architecture of 48 MaAHL proteins. (B) The conserved domain of 48 MaAHL proteins. (C) Exon-intron structure for each *MaAHL* gene. (D) The schematic diagram of three types of MaAHL proteins based on the AT-hook motifs and PPC/DUF296 domain. (E) The sequence structure of AT-hook1 (motif 3) and AT-hook2 (motif 5) in banana.

were intronsless or have only one intron except *MaAHL31* genes, suggesting MaAHL31 is not a typical type I AHL member, although it has a DUF296/PPC domain and a AT-hook1 motif. Introns are essential for maintaining pre-mRNA secondary structure and composing part of coding sequences through alternative splicing, thus the functional divergence of MaAHL members might be caused by the difference in intron numbers. The ATL proteins were reported to be localized in the nucleus as transcription factors (Street et al., 2008; Zhao et al., 2009; Wang et al., 2023). Subcellular localization prediction of MaAHL proteins showed that all of them were located in the nucleus (Table S1). In addition, chloroplast signaling peptides were identified in several MaAHL proteins, indicating these proteins may play a role in the plast development of banana.

Accumulating evidences revealed that the tissue-specific expression patterns of *AHL* genes are crucial for their regulation of plant growth and development mediated by phytohormones signaling. *AHL15* was expressed in the vascular cambium area and regulates cambial cell division activity in *Arabidopsis* (Rahimi et al., 2022a). AHL3 and AHL4, as direct intercellular signals by moving from one cell to another, regulating vascular tissue boundaries in *Arabidopsis* roots (Zhou et al., 2013). The *cis*-elements of promoters are generally essential in gene expression, and identifying them will be keys to exploring their potential roles in gene function. In this study, several elements involved in meristem expression, auxin and gibberellin responsiveness were found in the promotors of many *MaAHL* genes (Fig. 4), indicating *MaAHL* genes might also be participated in phytohormones-mediated plant development processes in banana. Jasmonic acid (JA) is a key signaling molecule in plant defense against biotic and abiotic stresses (Turner et al., 2002; Wasternack et al., 2006). AT-hook family members are involved in determining the level of expression of ORCA3 in response to jasmonate, involving in the jasmonate-responsive activation of terpenoid indole alkaloid biosynthetic genes in *Catharanthus roseus* (Vom et al., 2007). A number of MeJA-responsiveness elements have been identified in the promoters of the *MaAHL6* and *MaAHL9* genes, while these two genes

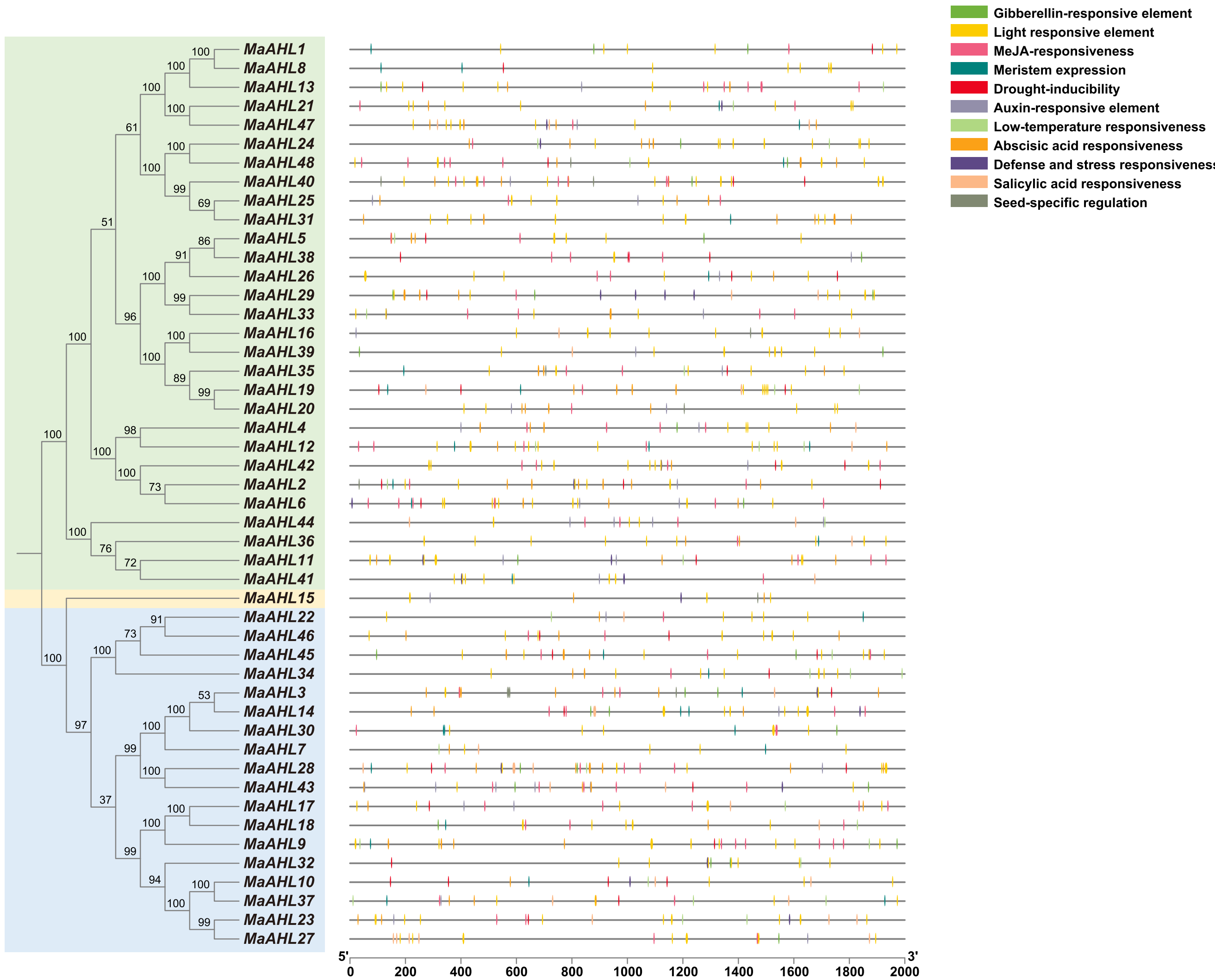

Fig. 4. Distribution of *cis*-elements in the promoter regions of *MaAHL* genes.

were significantly down-regulated in root of three banana genotypes under osmotic stress (Fig. 5), suggesting they might be a part of the JA-mediated plant defense response. The light responsive elements were identified in the promoter region of all MaAHL members. Analogously, all *AHL* gene promoters in grape, soybean and radish contained light response elements (Li et al., 2021; Wang et al., 2021; Chen et al., 2024). This suggests that the *AHL* genes in banana and other plant species could affect plant growth, development, and stress response mediated by light signals.

The characterization of fruit ripening is one of the important traits in banana. Interestingly, it was found that the expressions of clade-A *MaAHL* genes were generally downregulated while clade-B *MaAHL* genes were up-regulated during fruit ripening in banana, which further demonstrated the functional divergence of *MaAHL* genes (Fig. 6). The expression of *MaAHL43* gene was significantly up-regulated in middle and late stages of fruit ripening. The expression changes of ethylene synthesis related genes are the main indicators of banana fruit ripening (Inaba et al., 2007; Xiao et al., 2013). Overexpression of *MaAHL43* gene could activate the expression of ethylene synthesis related genes in advance, thereby promoting banana fruit ripening (Fig. 7). AHL family members could bind to AT-rich DNA regions in the form of hetero-multimeric complexes to act by inhibiting or activating target gene expression (Xu et al., 2013; Rahimi et al., 2022b). The MaAHL43 protein is located in nuclear and has no transcriptional activation activity in yeast cells, indicating MaAHL43 might be a transcriptional repressor or cofactor to involve in banana fruit ripening. Taken together, this study is the first to systematically characterize the AHL gene family in banana, providing a new target for studying the regulation network of banana fruit ripening.

4. Materials and methods

4.1. Identification and characterization of MaAHL genes

The sequences of 29 Arabidopsis AHL proteins were downloaded from TAIR database (https://www.arabidopsis.org/). The whole genome database of *M. acuminata* var. DH-Pahang (v4) was downloaded from Banana Genome Hub (https://banana-genome-hub.southgreen.fr/). To identify MaAHL proteins, a BLASTP search according AtAHL protein sequences were performed in *M. acuminata* genomes with E-value < 0.01. The potential sequences were submitted to the Pfam (http://pfam.sanger.ac.uk/search) and SMART (http://smart.embl-heidelberg.de/) to assess the AT-hook motif and PPC/DUF296 domains. The characterization of MaAHL members, including number of amino acids, the isoelectric point (pI), molecular weight (Mw) and grand average of hydropathicity (GRAVY), were calculated using the ProtParam tool of the ExPASy Server (https://web.expasy.org/protparam/). Subcellular localization of MaAHL proteins were predicted by Plant-mPLoc (http://www.csbio.sjtu.edu.cn/bioinf/plant-multi/).

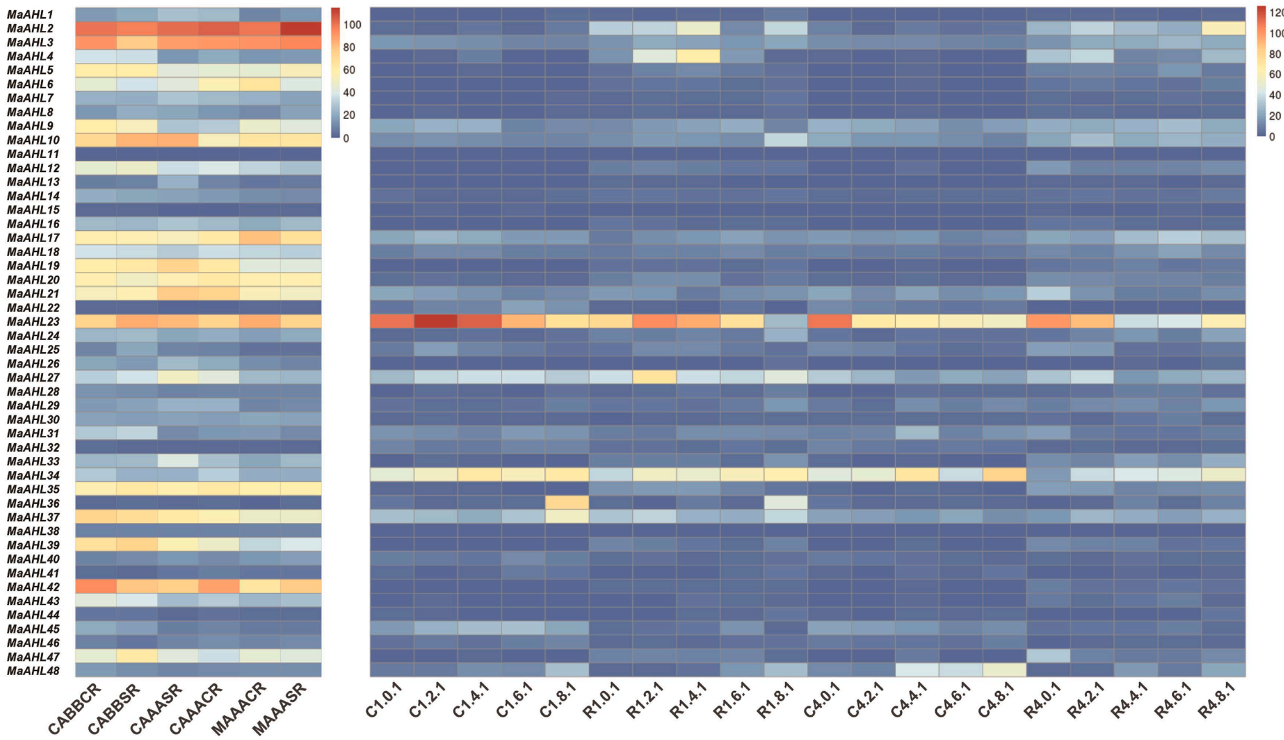

Fig. 5. The expression patterns of *MaAHL* genes under abiotic and biotic stress. (A) The heat map of *MaAHL* genes in root tissue of three genotypes under osmotic stress. CABBCR, GAAACR and MAAACR represent control root of three varieties, repectively. CABBSR, GAAASR and MAAASR represent stress root of three varieties, repectively. (B) The heat map of *MaAHL* genes in root and corm response to *Fusarium oxysporum* f.sp. *cubense* (*Foc*) race1 (R1) and tropical race4 (TR4). C1.0.1, C1.2.1, C1.4.1, C1.6.1 and C1.8.1 represent the corms of 0, 2, 4, 6 and 8 days after *Foc* R1 infection, respectively. R1.0.1, R1.2.1, R1.4.1, R1.6.1 and R1.8.1 represent the roots of 0, 2, 4, 6 and 8 days after *Foc* R1 infection, respectively. C4.0.1, C4.2.1, C4.4.1, C4.6.1 and C4.8.1 represent the corms of 0, 2, 4, 6 and 8 days after *Foc* TR4 infection, respectively. R4.0.1, R4.2.1, R4.4.1, R4.6.1 and R4.8.1 represent the roots of 0, 2, 4, 6 and 8 days after *Foc* TR4 infection, respectively.

4.2. Chromosomal localization and phylogenetic analysis of MaAHL proteins

The information about chromosome localization of *MaAHL* genes were obtained from the the GFF3 file of *M. acuminata* genome database and visualized using the Tbtools software (v2.0906) according to the manufacturer's instructions (Chen et al., 2020). For phylogenetic analysis, the multiple sequences alignments of AtAHL and MaAHL proteins were generated by Clustal W. Phylogenetic trees were constructed from multiple sequence alignments using the neighbor- joining method on the p-distance with pairwise deletion and 1000 bootstrap replicates in MEGA 6 (Kumar et al., 2016). The phylogenetic tree was further visualized using Evolview (http://www.evolgenius.info/evolview/) (Subramanian et al., 2019).

4.3. Conserved motif, domain and gene structure analysis of MaAHL proteins

The Multiple Expectation maximizations for Motif Elicitation (MEME) program (ver. 5.1.1, http://meme-suite.org/) was employed to identify the potential conserved motifs of the MaAHL proteins. The conserved domain of MaAHL members were further confirmed by CD-search in the NCBI database (https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi). The information about gene structure for *MaAHL* genes were obtained from the GFF3 file of *M. acuminata* genome database. Gene structure and conserved motif within the phylogenetic tree of MaAHL proteins were visualized by TBtools software (v2.0906) according to the manufacturer's instructions (Chen et al., 2020).

4.4. Cis-element analysis of MaAHL genes promoters

The upstream 2 kb sequence of transcription start site (TSS) for each *MaAHL* gene was extract by TBtools software as the presumptive promoter. The promoter sequences of *MaAHL* genes were loaded into the PlantCARE (http://bioinformatics.psb.ugent.be/webtools/plantcare/html/) database to identify potential *cis*-element and further visualized by TBtools software (v2.0906).

4.5. Transcriptome-based expression profiling of MaAHL genes

The previous transcriptome data, including root tissue under osmotic stress, the root and corm response to *Fusarium oxysporum* f.sp. *cubense* (*Foc*) race1 (R1) and tropical race4 (TR4), were employed to explored the expression profiles of *MaAHL* genes (Zorrilla-Fontanesi et al., 2016; Anuradha et al., 2024). Heat maps were generated by the Banana Genome Hub (https://banana-genome-hub.southgreen.fr/) with the log count per million.

4.6. RNA sequencing

The harvested fruits of genotype 'Jiali' (*M. acuminata*, AA group) were ripened at 25°C for 5 days in closed polyethylene bags. The fruit at different stage P1 (1 day after harvest), P3 (3 days after harvest) and P5 (5 days after harvest) were collected for transcriptome analysis. Two biological replicates were used for transcriptome analysis. Total RNA was extracted by Trizol reagent kit (Invitrogen, Carlsbad, CA, USA) following the manufacturer's instructions. The integrity, purity and concentration of total RNA were detected by RNase free agarose gel electrophoresis and Agilent 2100 Bioanalyzer (Agilent Technologies, Palo Alto, CA, USA). After total RNA was extracted, eukaryotic mRNA

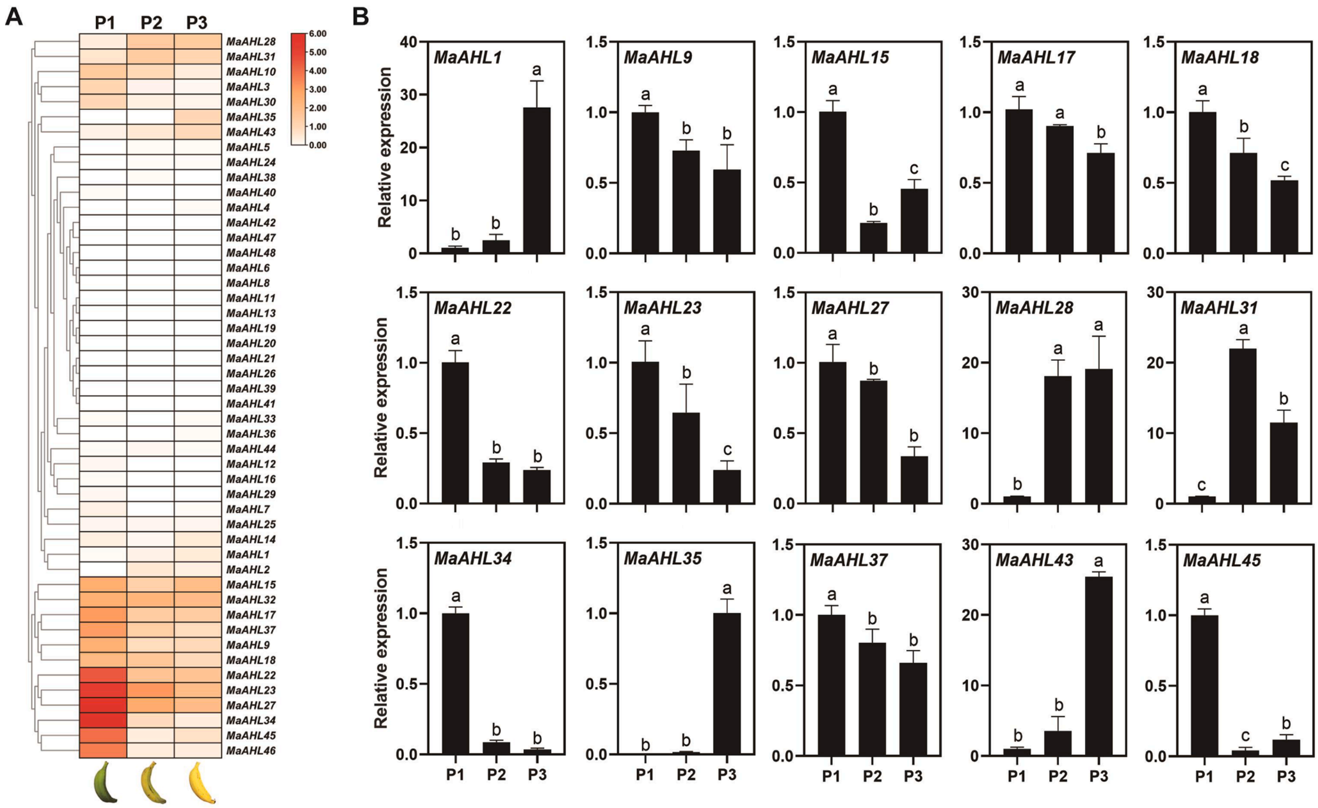

Fig. 6. The expression profiles of *MaAHL* genes during fruit ripening. (A) The heat map of all *MaAHL* genes expression pattern in fruit at three different ripening stages. (B) Validation of 15 *MaAHL* genes expression data from RNA-Seq by RT-qPCR. Different red letters indicate significant differences at level *p* value = 0.05.

was enriched by Oligo(dT) beads, then the enriched mRNA was fragmented into short fragments using fragmentation buffer and reversly transcribed into cDNA by using NEB Next Ultra RNA Library Prep Kit for Illumina (NEB #7530, New England Biolabs, Ipswich, MA, USA). The purified double-stranded cDNA fragments were end repaired, A base added, and ligated to Illumina sequencing adapters. The ligation reaction was purified with the AMPure XP Beads (1.0X), and polymerase chain reaction (PCR) amplified. The resulting cDNA library was sequenced using Illumina Novaseq6000 by Gene Denovo Biotechnology Co. (Guangzhou, China). Genes with FDR < 0.05 & $|\log_2(\text{foldchange})| \geq 1$ found by DESeq2 were assigned as differentially expressed. Heatmap for expression profiles of *MaAHL* genes were generated with TBtools software (v2.0906).

4.7. Real time quantitative polymerase chain reaction (RT-qPCR)

The qualified RNA samples were reverse-transcript into cDNA using a HiScript III 1st Strand cDNA Synthesis Kit (+gDNA wiper) (Vazyme Biotechnology Co., Nanjing, China) following the manufacturer's instructions. RT-qPCR analysis was performed in a BioRad CFX96 Real-Time PCR platform (BioRad, USA). The specific program parameters were set according to the manufacturer's instructions of Taq Pro Universal SYBR qPCR Master Mix (Vazyme Biotechnology Co., Nanjing, China). Three biological replicates and three parallel experiments were carried out. The relative expression levels were calculated by the $2^{-\Delta\Delta C}_{T}$ method with *RIBOSOMAL PROTEIN S* (*RPS*) gene as internal references (Chen et al., 2011). The primer sequence information for RT-qPCR analysis was displayed in Table S2.

4.8. Subcellular localization

Subcellular localization assay was performed in *Nicotiana benthamiana* leaves as previously described (Dong et al., 2023). The coding sequences of *MaAHL43* without the stop codon was cloned into the pCAMBIA1300-GFP vector to generate *35S:: MaAHL43-GFP*. The 35S:: GFP as a negative control, the *35S::H2B-mCherry* was used as a nuclear marker. The constructs were transformed into *Agrobacterium* strain GV3101. The recombinant constructs and the nucleus-localized marker were co-infiltrated in *N. benthamiana* leaves by *A. tumefaciens* GV3101. The infiltrated plants were cultured in the dark for 24 h, then under a 16 h light/8 h dark photoperiod for 2 days. The GFP and mCherry signal was detected using the confocal laser-scanning microscope (LSM800, Zeiss, Germany). The primer sequence information for vector construction was displayed in Table S2.

4.9. Transactivation activity assay

Transactivation activity assay was performed in yeast cells as previously described (Dong et al., 2023). The CDS of *MaAHL43* was cloned into the pGBKT7 to generate corresponding construct. Then, recombinant was introduced into yeast strain Y2HGold and the yeast transformants were plated and grown on SD-Trp and SD-Trp/His/Ade medium at 30 °C for 3 days. Empty pGBKT7 was used as a negative control, GAL4 was used as a positive control. The primer sequence information for vector construction was displayed in Table S2.

4.10. Transient overexpression in banana fruit

Transient overexpression was performed in banana fruit as previously described (Wei et al., 2023). The full-length CDS of *MaAHL43* was cloned into the pCAMBIA1300-GFP vector to produce the MaAHL43-OE construct. The constructs were transformed into *Agrobacterium* strain GV3101. The pulp of mature-green banana fruit of genotype 'Baxijiao' (*Musa* spp. AAA) was injected with *Agrobacterium* harboring the constructed plasmids. Then transformed fruits were treated with 100 μL·L^{-1} ethylene and held at 23 °C for 5 days. The pCAMBIA1300-GFP vector as

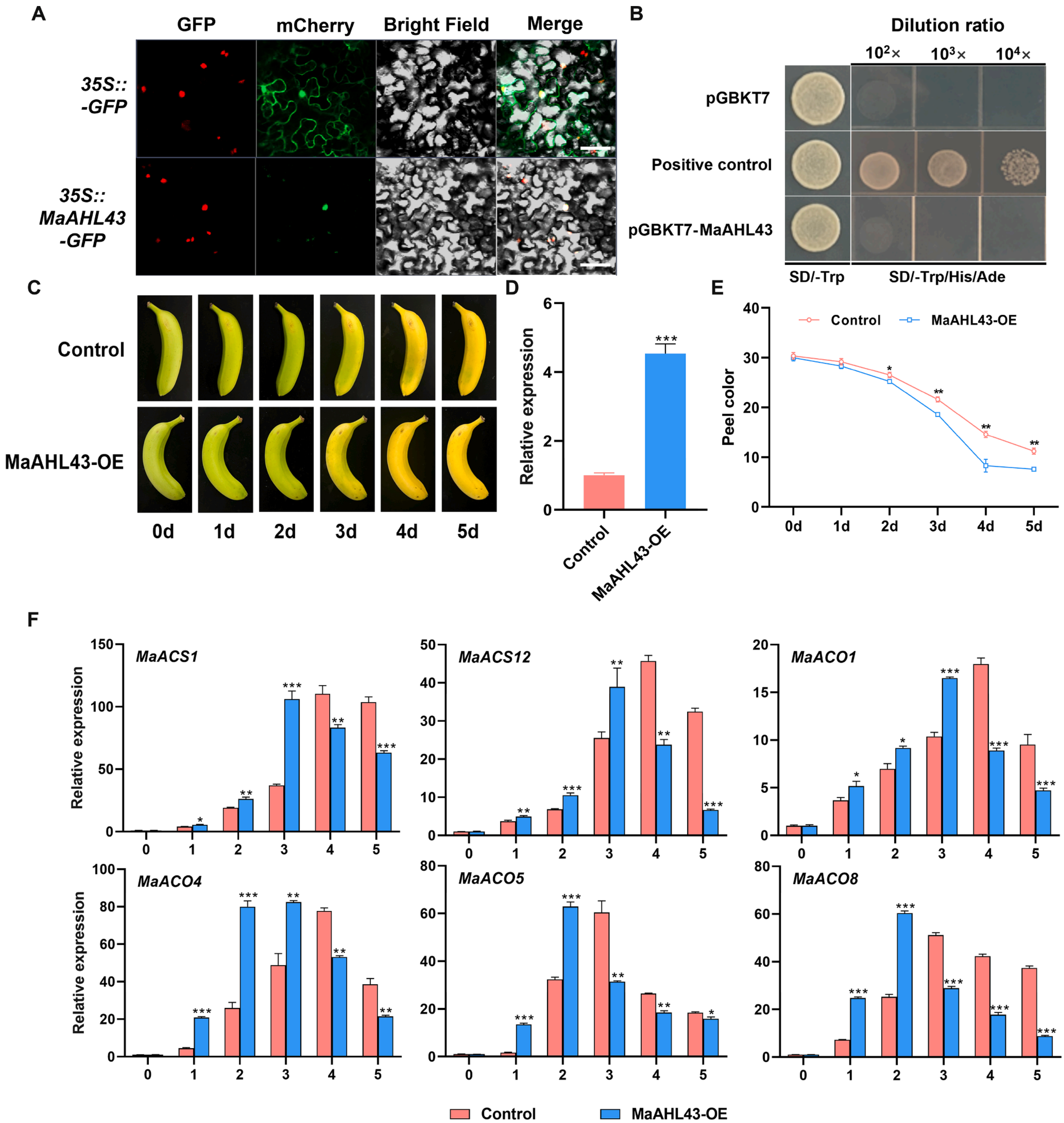

Fig. 7. Overexpression of *MaAHL43* gene could promote banana fruit ripening. (A) Subcellular localization of MaAHL43 protein in tobacco leaves. The plasmids *35S::GFP* and *35S::H2B-mCherry* were used as the negative control and nuclear marker, respectively. Bar = 50 μm. (B) Transactivation activity assay of MaAHL43 protein in yeast cells. Empty pGBKT7 was used as a negative control, GAL4 was used as a positive control. (C) The phenotype of banana fruit transiently overexpressed with MaAHL43-OE and empty vector during ripening. (D) The expression level of *MaAHL43* in transiently overexpressed and control banana fruit. (E) The peel color of transiently overexpressed and control banana fruit. (F) The expression levels of ethylene synthesis related genes in transiently overexpressed and control banana fruit.

a control. Three biological replicates and three parallel experiments were carried out. The peel color was measured using a spectrophotometer Konicaminolta CM-2300d.

4.11. Statistical analysis

Data were analyzed using SPSS statistical software (SPSS 20.0, IBM). Student's t-test was used to determine statistical significance between two groups. One-way analysis of variance with least significant difference (LSD) tests was used to determine statistical significance among multiple range test.

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Data availability

The raw sequence data reported in this paper have been deposited in the Genome Sequence Archive (Chen et al., 2021) in National Genomics Data Center (CNCB-NGDC Members and Partners, 2022), China National Center for Bioinformation / Beijing Institute of Genomics, Chinese Academy of Sciences (GSA: CRA018230) that are publicly accessible at https://ngdc.cncb.ac.cn/gsa.

CRediT authorship contribution statement

Junhui Dong: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Linbing Xu:** Writing – review & editing, Supervision. **Hongyun Zeng:** Writing – review & editing, Validation, Supervision. **Xingyu Yang:** Writing – review & editing, Data curation. **Yuanli Wu:** Writing – review & editing, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scienta.2024.113948.

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